

The 2% Illusion in Genomic Selection

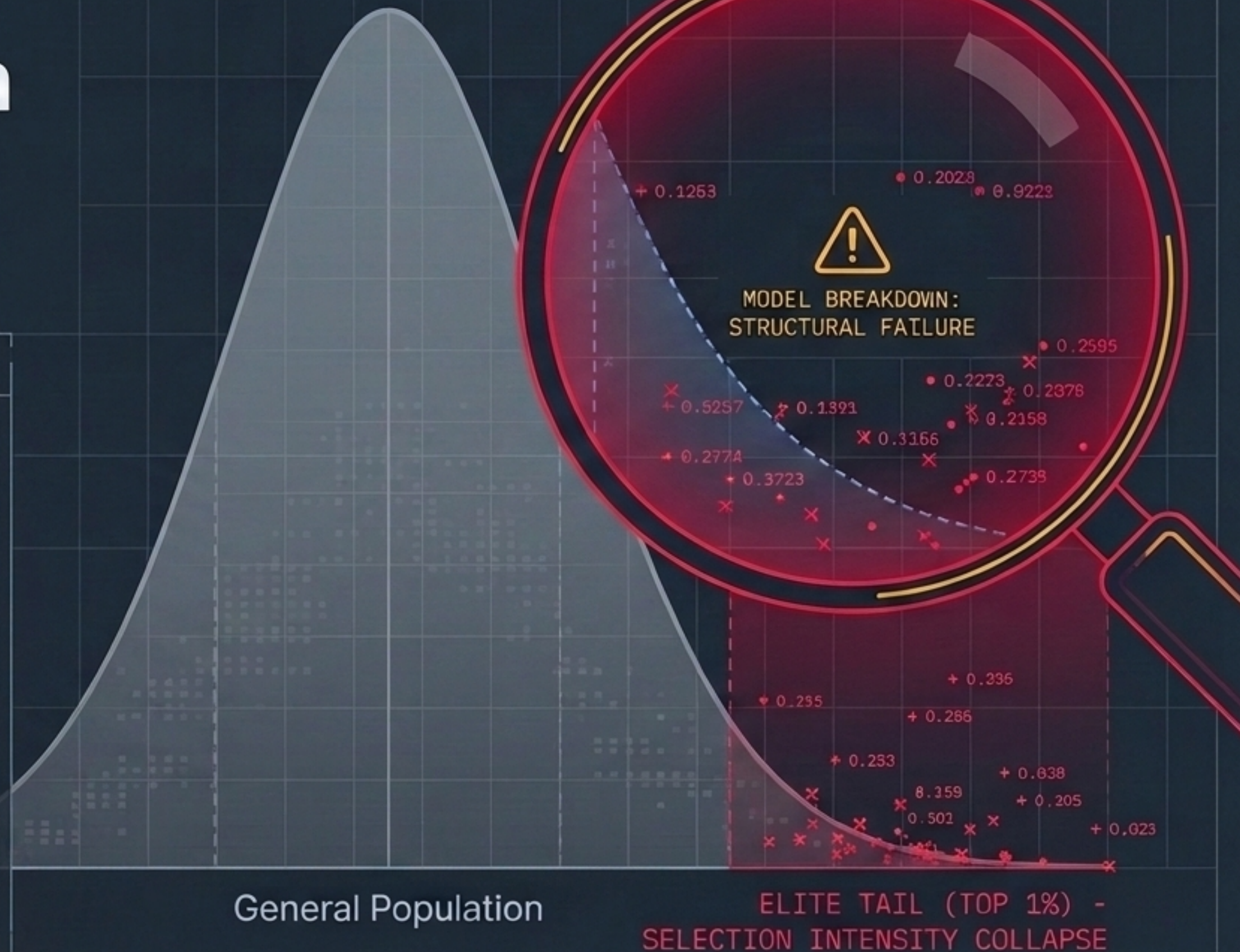
Exact G-BLUP validation reveals severe structural failure in APY approximations at the elite selection tail.

DIAGNOSTIC DATA DOSSIER

- Target Scope: **$N > 10,000$** genomic selection programs.
- The Subject: The **Algorithm for Proven and Young (APY)** [1].
- The Finding: The accepted **~2%** global accuracy loss [1] conceals an up to **85.6% collapse** in effective selection intensity at the **top 1% margin** [2].

References:

- [1] Misztal, I. et al. (2014). Implementation of single-step GBLUP... J. Dairy Sci.
[2] Fragomeni, B. O. et al. (2015). Use of genomic recursions... J. Anim. Sci.



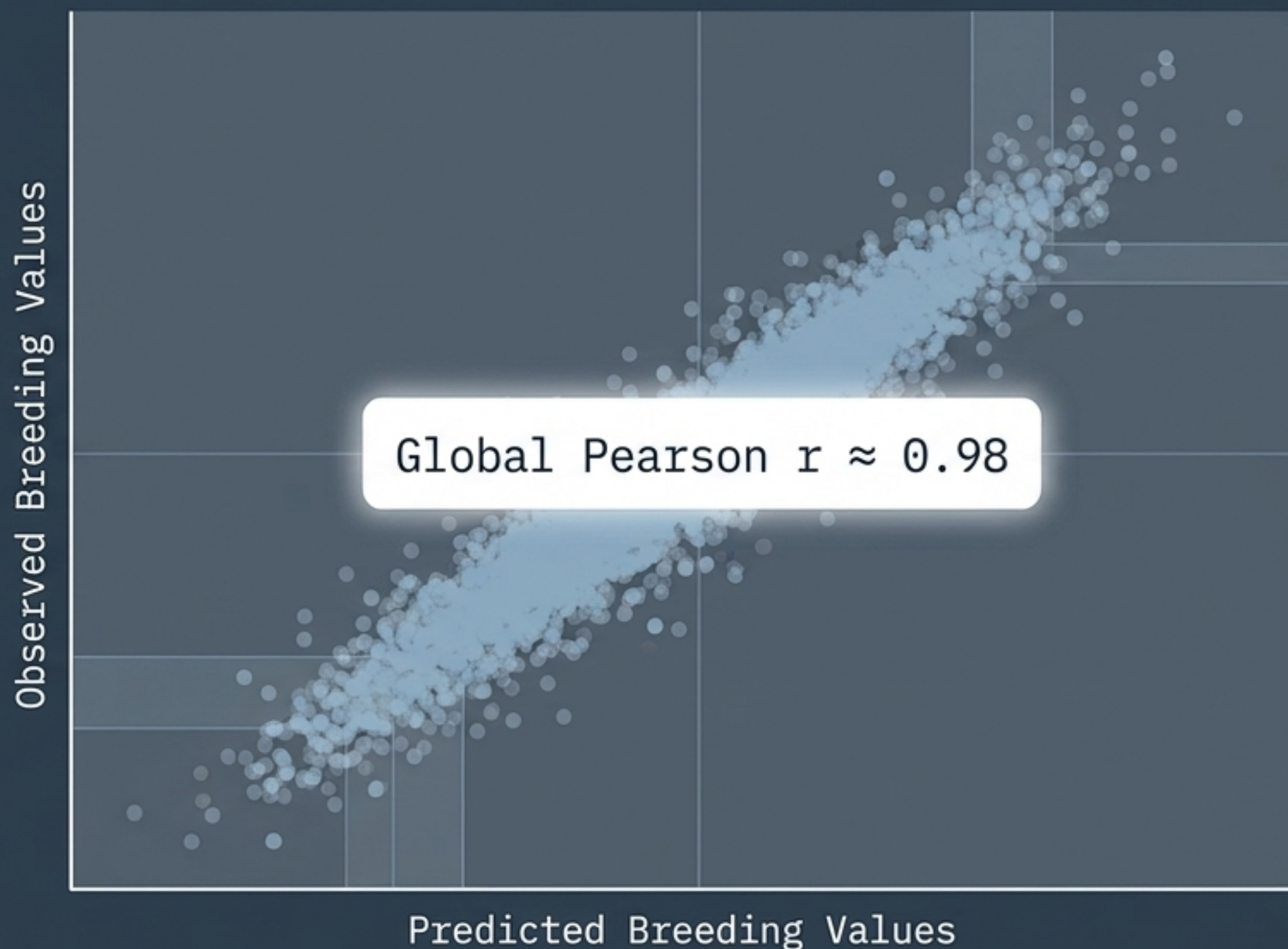
The $O(N^3)$ computational wall forces a costly compromise

The Bottleneck Reality Check			
Panel Size (N)	GRM Storage	CPU Exact Solve Time	Practical Status
1,000	8 MB	<1 s	Trivial
10,000	800 MB	~90 s	Slow but feasible
20,000	3.2 GB	~15 min	Painful
50,000	20 GB	~40 min	Infeasible per-cycle
500,000	2 TB	Months	Impossible

The APY
Threshold

- Genomic BLUP requires solving a dense, symmetric positive semi-definite matrix.
- Above $N=10,000$, exact $O(N^3)$ factorization becomes a bottleneck.
- The industry adopted APY: inverting only a small 'core' subset ($n_{\text{core}} \ll N$) to approximate the inverse.

The widely accepted myth of a 'safe' 2% accuracy penalty



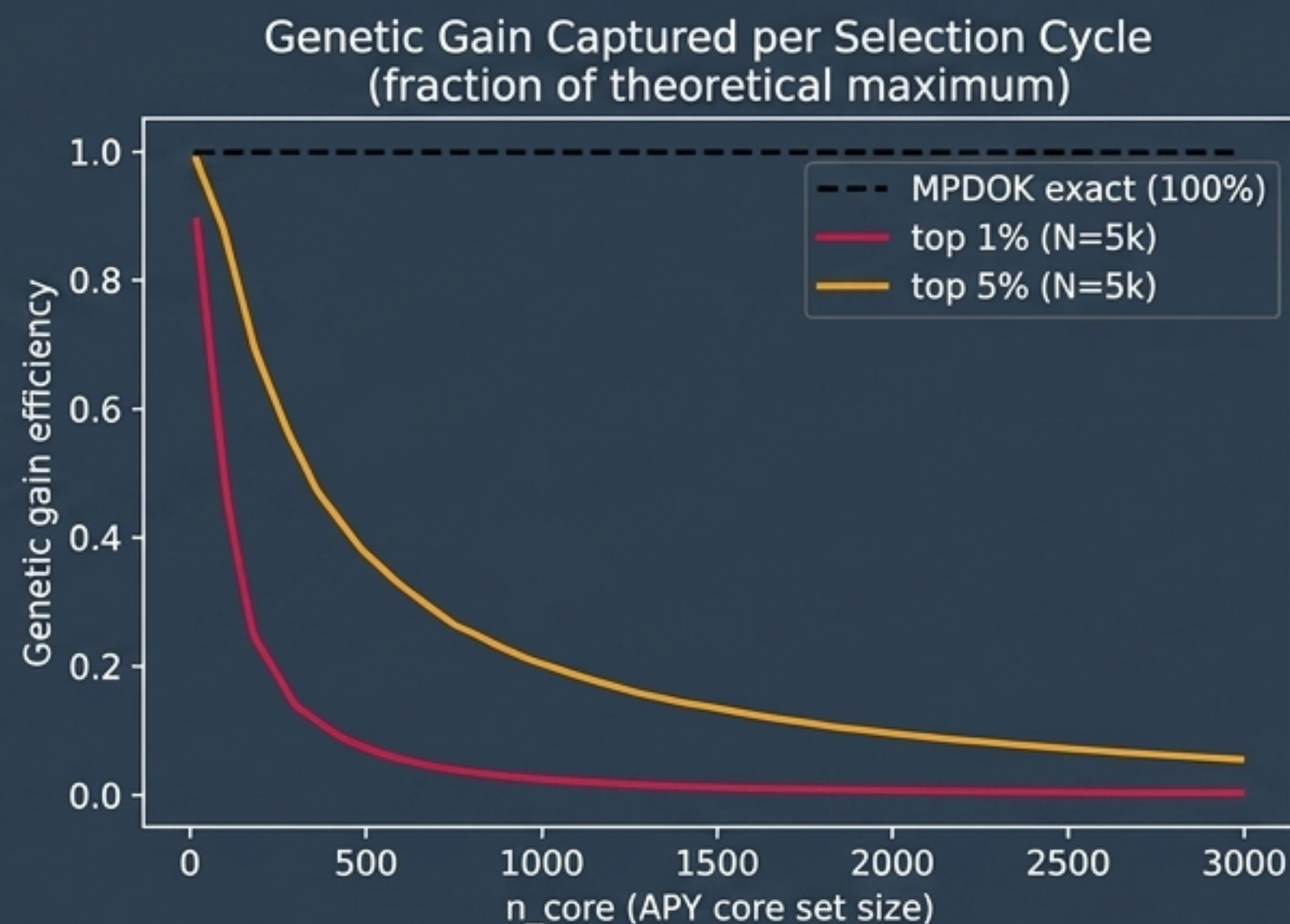
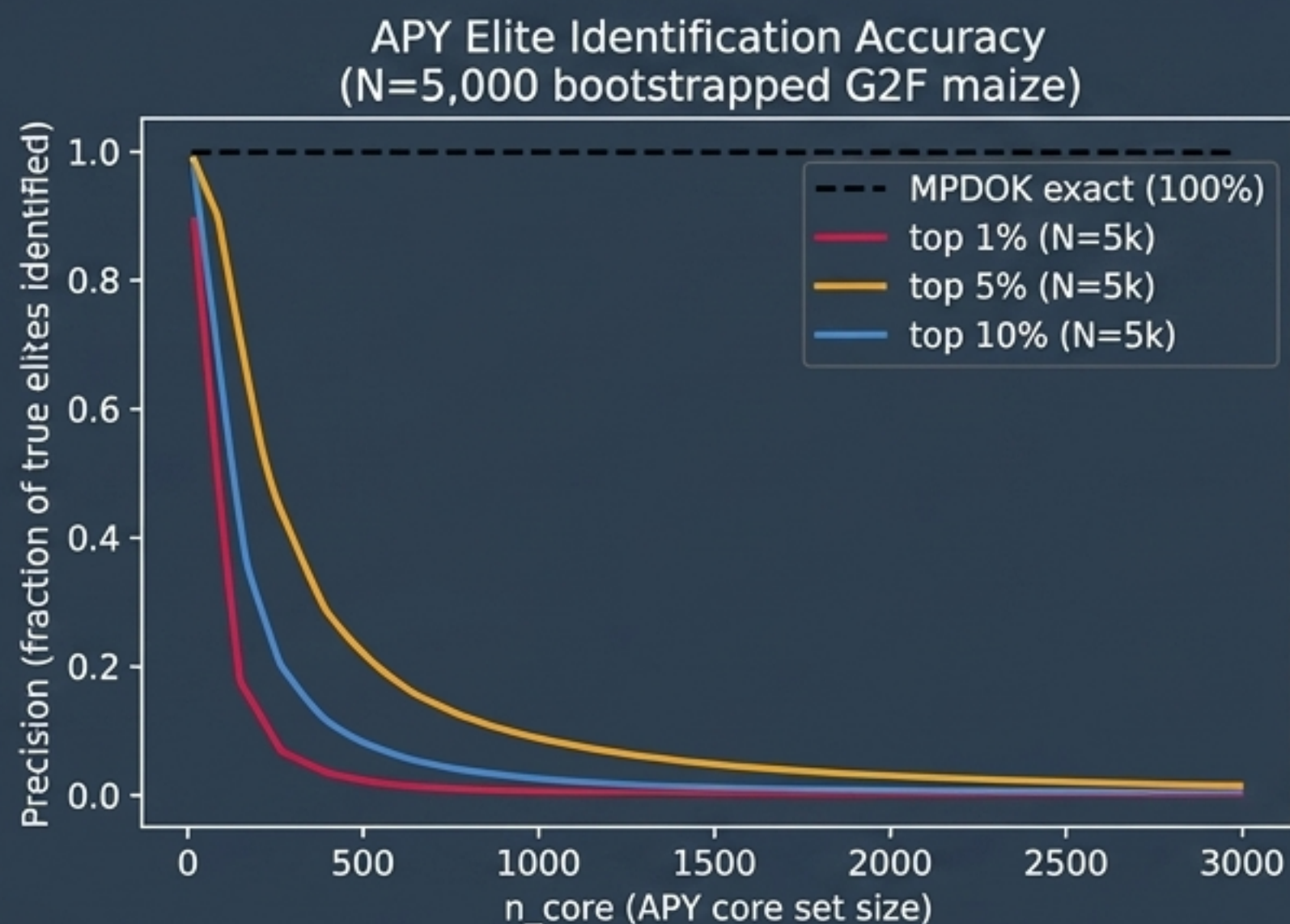
- The standard justification for APY is that it preserves nearly all global prediction accuracy.
- Across the entire population of N animals, the Pearson correlation (r) between the exact G-BLUP solution and the APY approximation typically drops by only 0.5% to 2%.
- The assumption: If $r \approx 0.98$, then genetic gain (ΔG) must remain at $\approx 98\%$ of the theoretical maximum.

Global averages camouflage catastrophic failure at the margins



- Breeding programs do not select the average. Economic value is entirely concentrated in the extreme right tail of the distribution.
- We audited APY's performance strictly at the elite margin using real G2F maize genomic sequencing data (N=2,193 lines, 48,580 real SNPs).
- The discovery: APY's α coefficient relative error is 16x to 41x worse than the exact solution, fundamentally misranking the true elites.

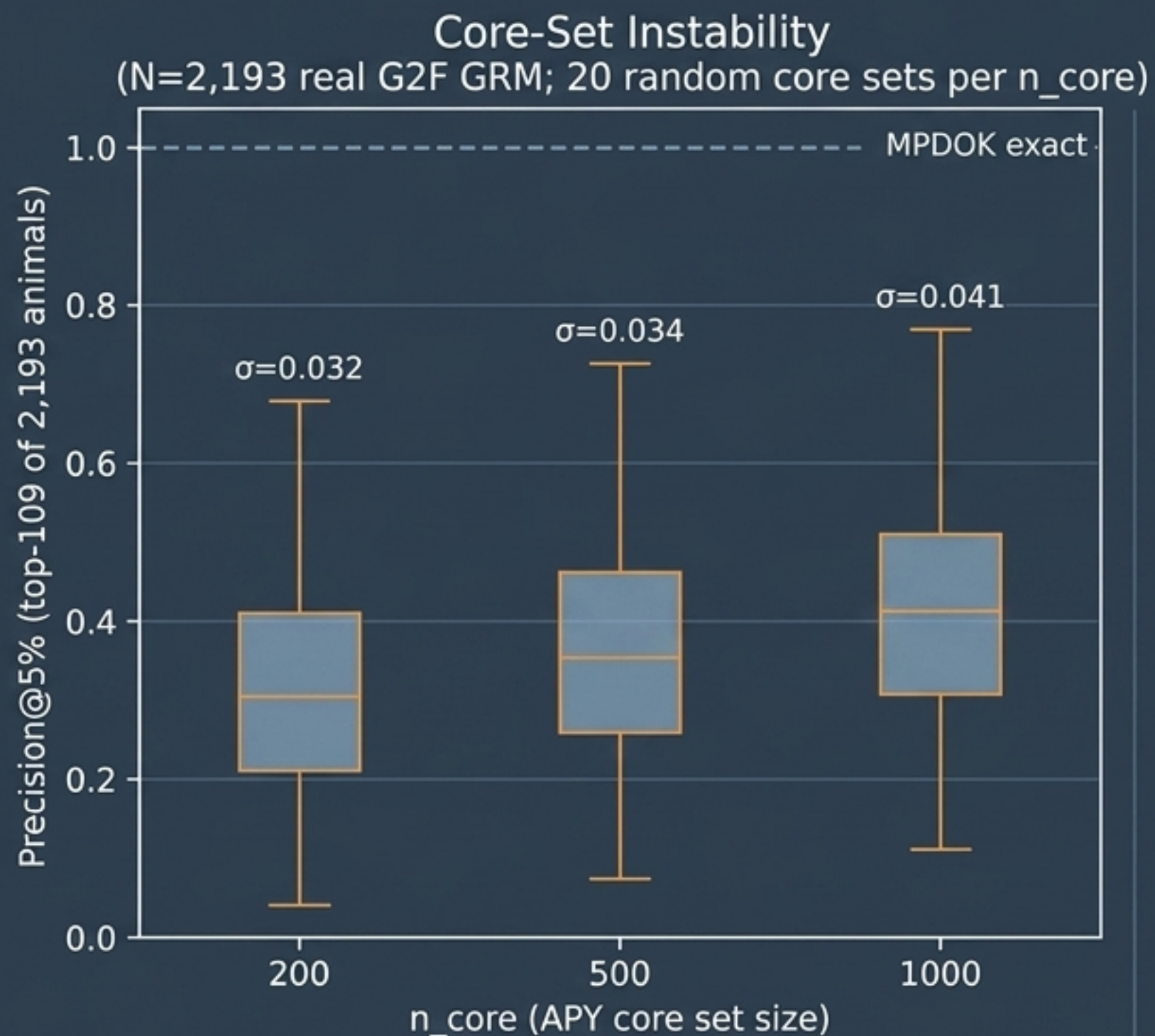
Empirical proof: APY misses up to 95% of true elite candidates



At the industry minimum ($n_{\text{core}} = 200$), APY correctly identifies only 1 animal out of 21 true top-1% candidates (4.8% precision).

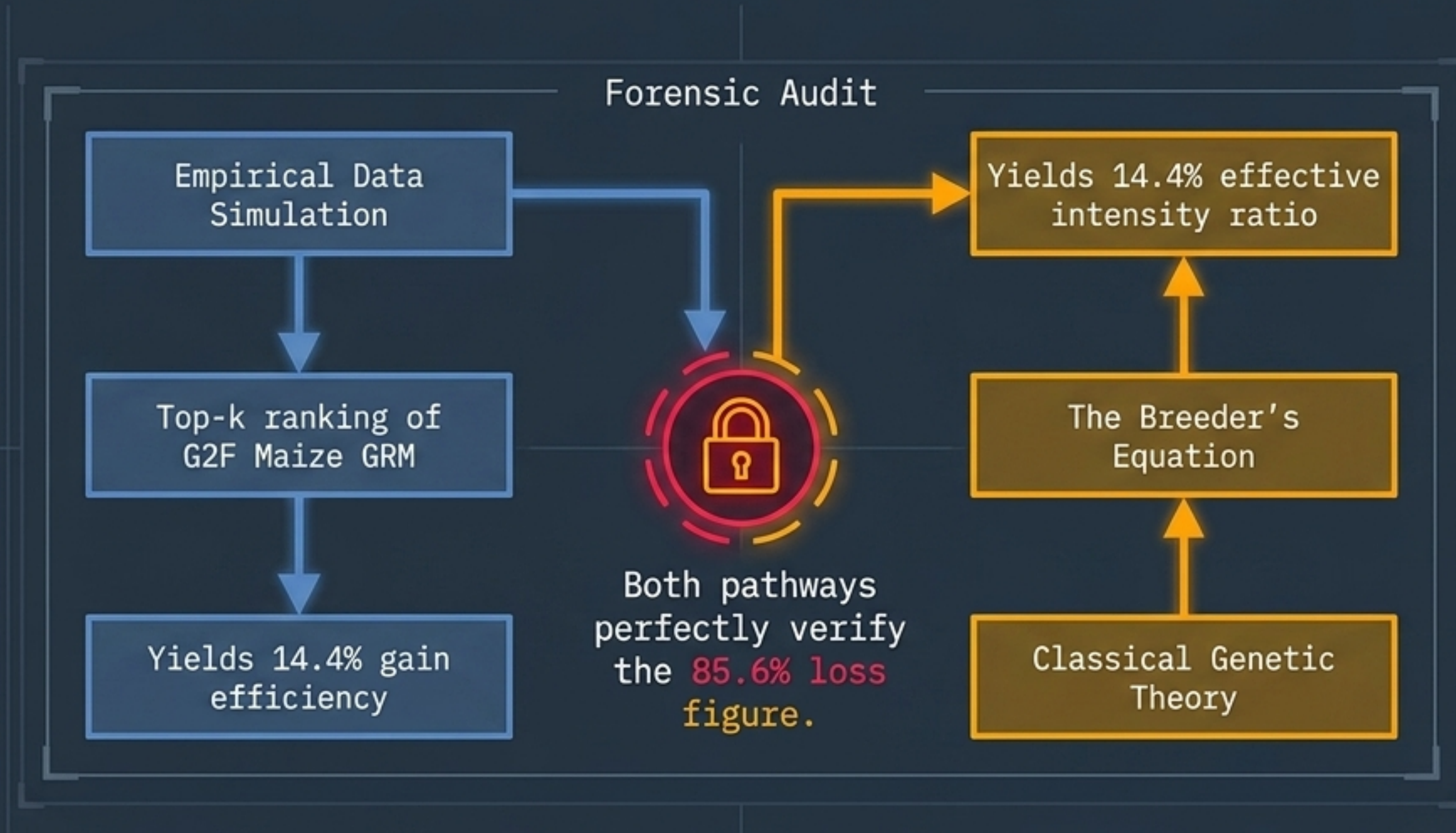
Genetic Gain Efficiency: Selecting the top-1% by APY rank captures only 14.4% of the maximum possible true genetic gain. The program captures just 14 cents of every dollar of potential genetic progress per cycle.

The core-set lottery introduces severe instability



- APY results depend entirely on which specific animals are randomly designated as the core.
- At N=2,193 (n_core=200), precision at the top-5% swings wildly between **11.0%** (worst case) and **22.9%** (best case).
- The implication: Two independent laboratories utilizing the exact same genomic data, but initializing with different random core sets, **will disagree on which animals to select by a factor of two.**

Rigorous two-way verification isolates the exact mechanism of failure



- Because these findings challenge foundational industry practices, they require dual-vector validation.
- We cross-checked the observed empirical collapse in precision against the classical Breeder's Equation to identify how the model fails globally.

The Breeder's Equation reveals the 86% collapse in selection intensity

$$\Delta G = \frac{i \cdot r \cdot \sigma_A}{L}$$

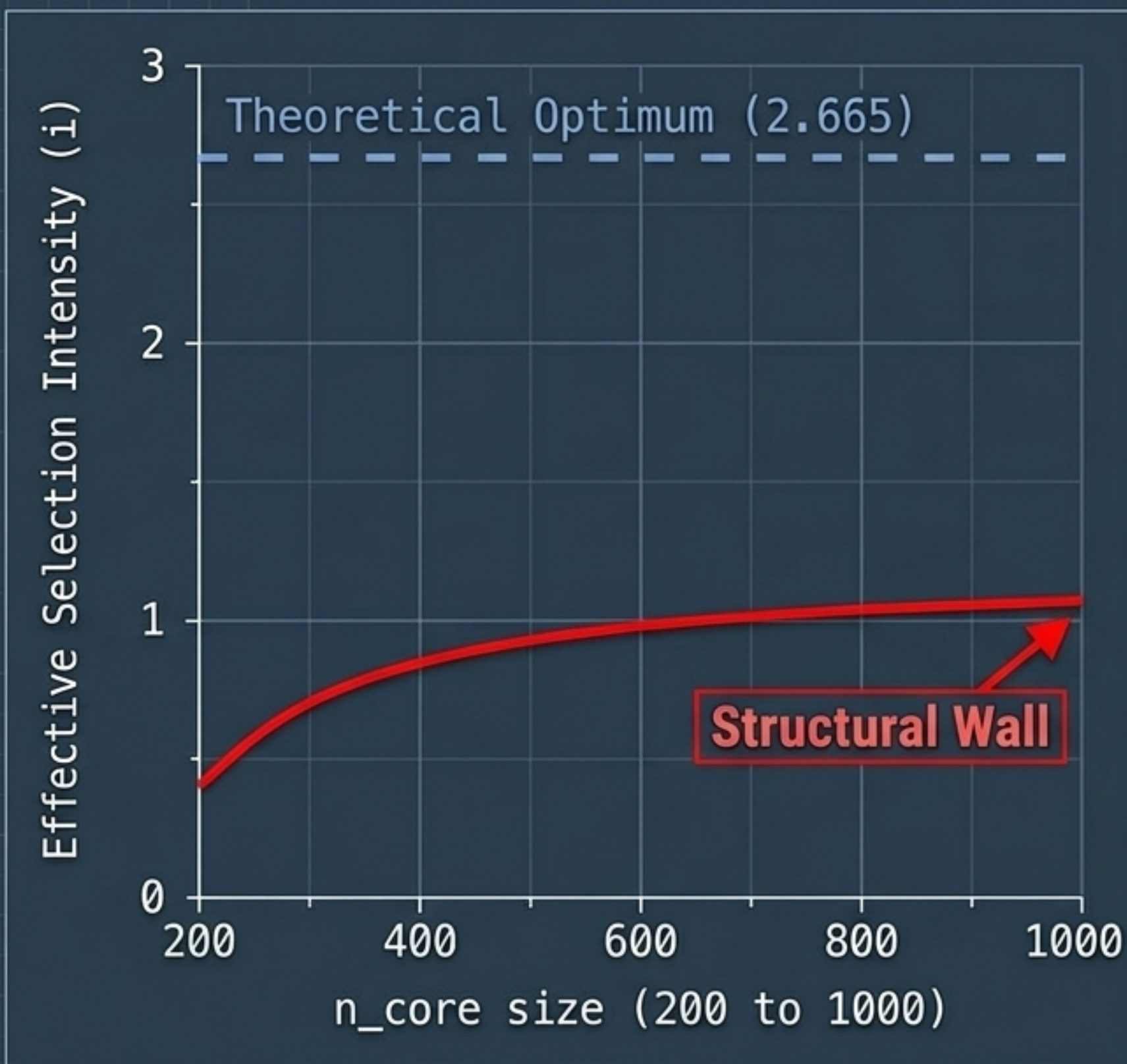


APY preserves global correlation $r \approx 0.98$.

APY guts elite identification.

Math stats	Insight
Theoretical i (Top 1%): 2.665	The "2% r " logic collapses distinct variables . APY ranks are so distorted at the tail that selecting the APY ' Top 1% ' is mathematically equivalent to drawing randomly from the top 78% of the population .
APY Effective i ($n_{\text{core}}=200$): 0.384 (-85.6% reduction)	

The failure of the “large core” safety net



The standard response to APY criticism is to massively increase the core set. **The math proves this fails.**

Increasing n_{core} from 200 to 1,000 requires 5x more animals (consuming 46% of the $N=2,193$ reference population).

The Payoff: Effective i rises to 1.026, which is still 61.5% below the theoretical optimum.

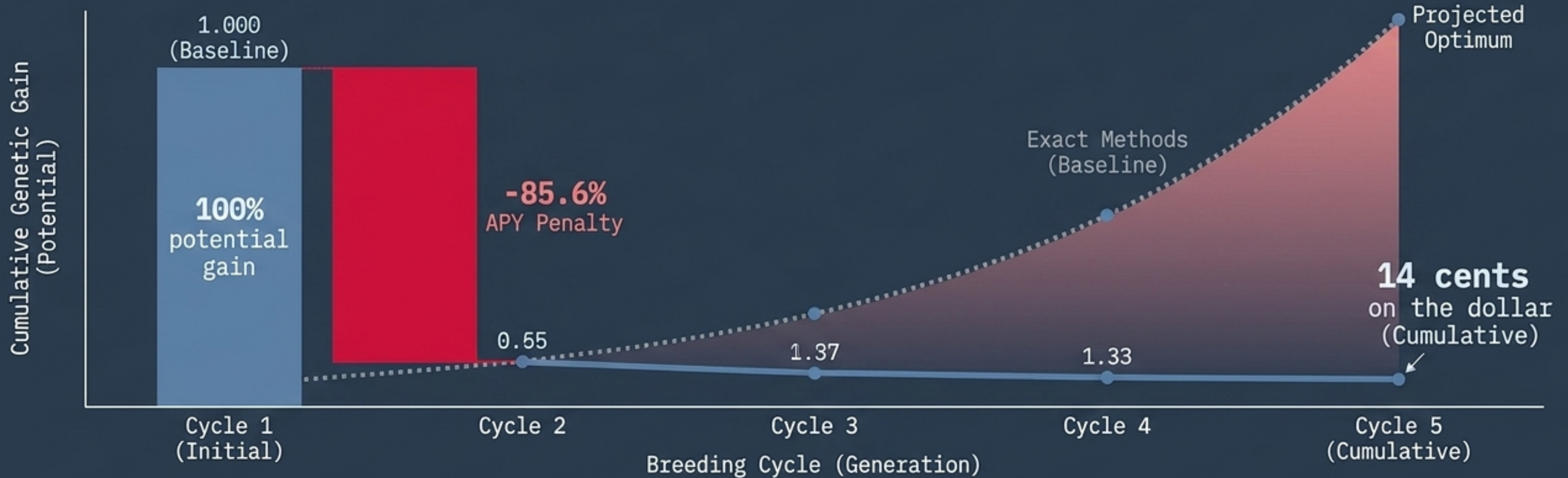
This is a bounded-rank matrix constraint: you cannot compute your way out of the penalty by scaling the core.

The Illusion vs. Reality Matrix

The Accepted APY Myth	The Verified Reality
Global Accuracy (r) High (~ 0.98)	High (~ 0.98), but globally irrelevant.
Elite Identification Minor penalty	APY misses up to 95% of true Top-1% elites.
Selection Intensity (i) Preserved	Collapses by 85.6% per cycle.
Core-Set Stability Stable baseline	$\pm 3.2\%$ to 4.1% standard deviation; highly volatile.

Unseen approximation penalties compound across generations

Predictive Waterfall Chart

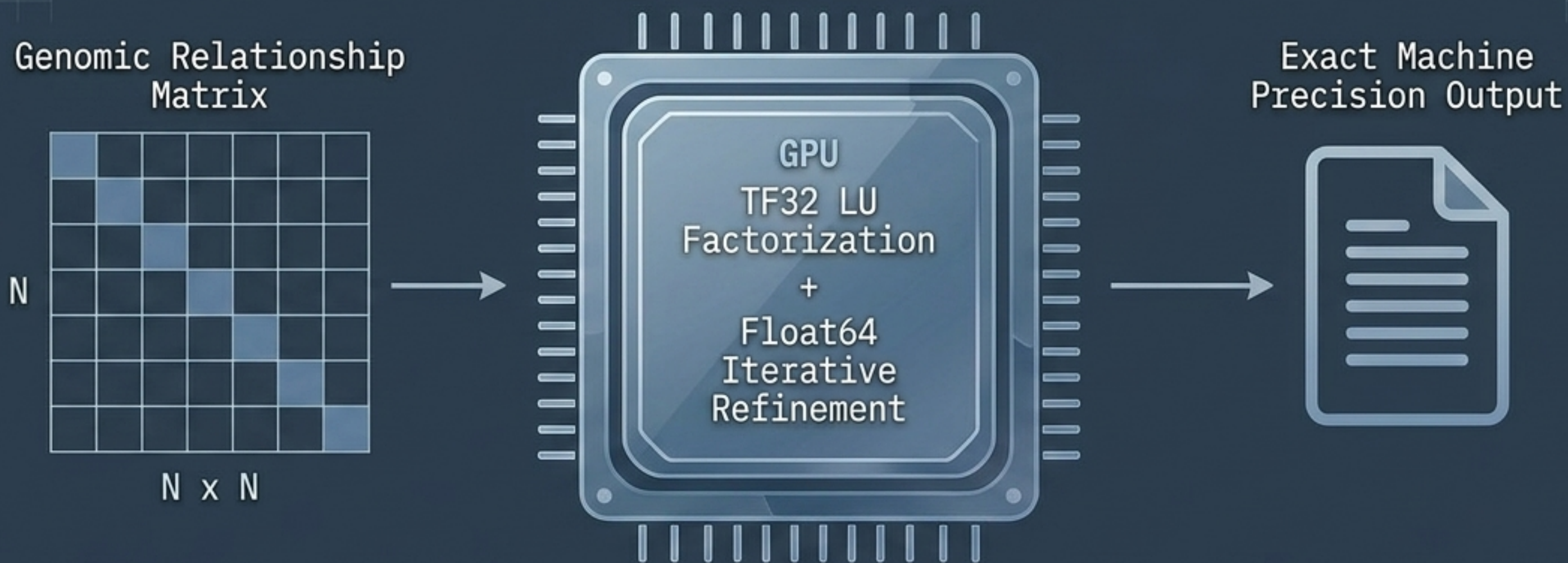


A **1% accuracy improvement** in a major breeding program translates to **millions of dollars**.

Bearing the full operational cost of elite selection (strict culling, generation-interval pressure) while capturing the genetic gain of a near-population-average draw **destroys commercial value**.

Note: Because these findings disrupt standard practice, we strongly invite **independent external validation** using the provided open-source datasets.

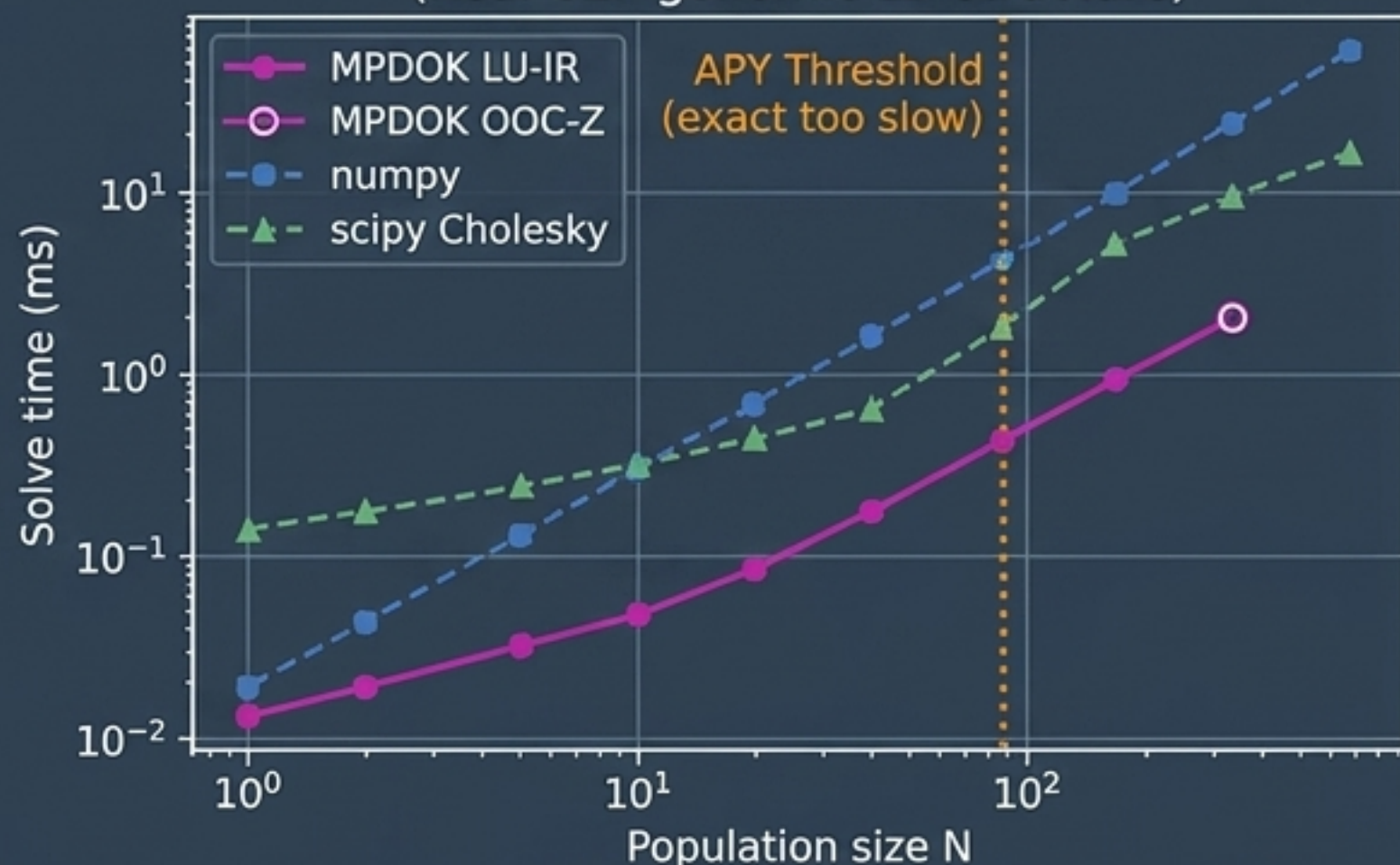
The Solution: Exact dense linear solves at scale via MPDOK



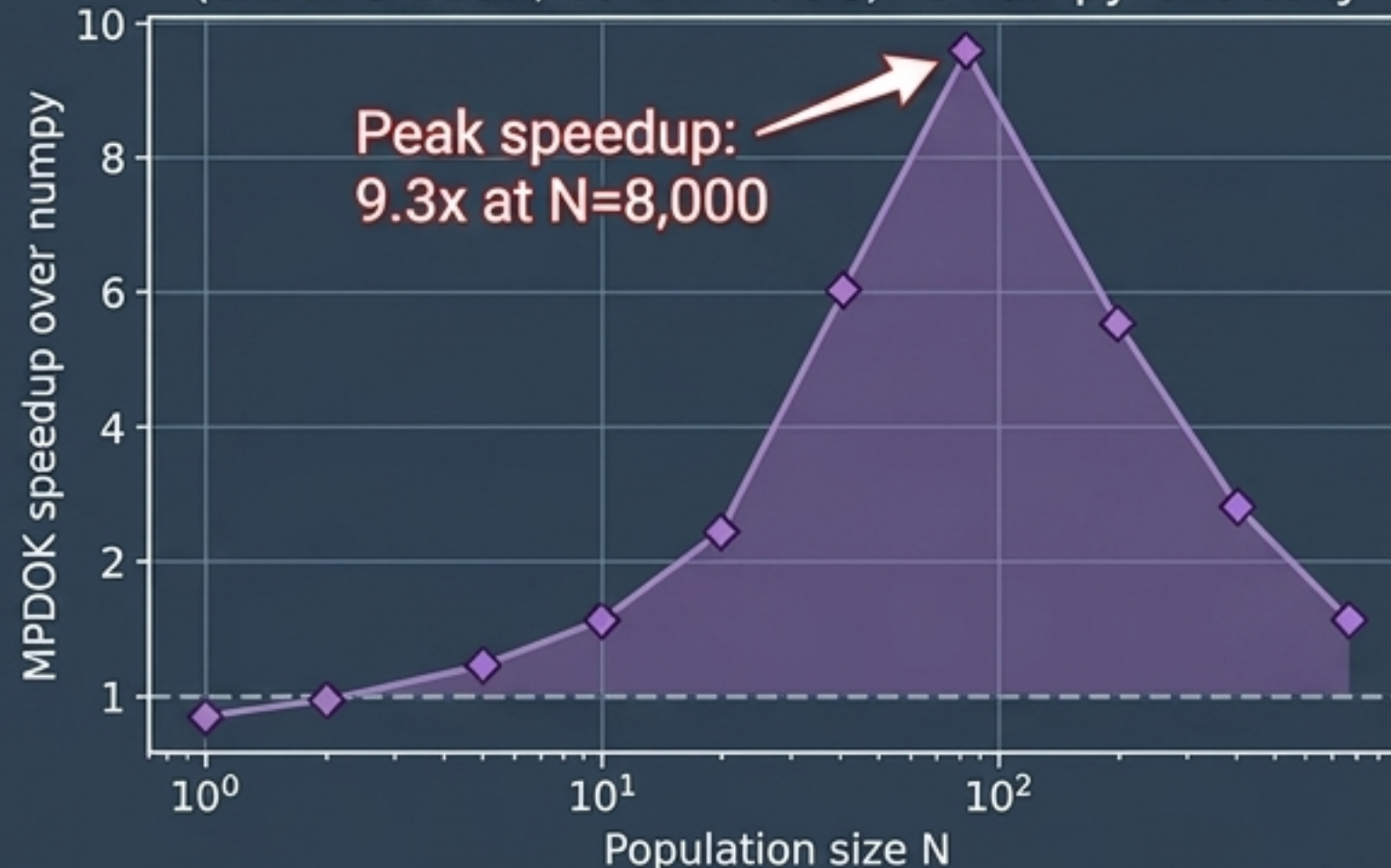
- MPDOK (Mixed Precision Dense OK) entirely **removes the $O(N^3)$ computational bottleneck**.
- Dense, symmetric positive semi-definite GRMs are **perfectly matched** for MPDOK's TF32 factorization + FP64 refinement.
- **Zero approximation penalty**: Captures **100%** of the elite tail. Matches numpy/scipy to machine precision (**$1.43e-15$**).
- At **$N=2,193$** , MPDOK delivers the **exact solution** in **86 ms** (vs APY approximation at **2.4 s**).
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Pushing the exact G-BLUP frontier beyond the APY threshold

G-BLUP Solve Time vs Population Size
(Real G2F genomic LD structure)



MPDOK Speedup
(exact G-BLUP, LU-IR + OOC) vs numpy Cholesky



MPDOK LU-IR Mode: Solves $N=10,000$ exactly in 1.28 seconds (2.5x faster than CPU). Peak 9.3x speedup at $N=8k$.

MPDOK OOC-Z Mode: Out-of-core solver entirely avoids PCIe transfer bottlenecks by computing on-the-fly inside GPU HBM.

Crosses $N=30,000$ in 53.7 seconds (1.1x faster than numpy), achieving exact solutions where industry currently gives up and relies on APY.

The Elite Identification Matrix: Rescuing Selection Intensity

Metric	APY (n_core=200)	APY (n_core=1,000)	Exact MPDOK
Top-1% Precision	4.8%	9.5%	100%
Top-1% Effective i	0.384	1.026	2.665
Core-Set Instability	$\pm 3.2\%$	$\pm 4.1\%$	0%
Solve Time	2.4 s	2.4 s	86 ms

Bottom Line: APY is an avoidable accuracy loss. MPDOK rescues selection intensity at the commercial tail where genetic gain is won or lost.